True Terminal Activation on Canonical Multiscale Incidences: Pair-Lift Minimality, Parent Inflation, and Sharp Capacity Laws

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Abstract

TPC–63 separates a literal full incidence from its pair projection and shows that a labeled multiscale completion is an isometry only after the completed incidence has been chosen. It leaves open a prior arrow: does a pair-populated or multiscale repair preserve the true pre-terminal parent, and what terminal lower constant remains after any added parent mass is charged?

We answer that finite-scale representation question at one fixed nonzero shift. On blockwise full incidences we define the literal parent mass P_m, terminal-support mass H_m, and shaped terminal energy T_m. The least parent-complete pair relation is the projection of the literal pre-terminal active incidence. Such a relation is also terminal-atomically faithful if and only if its terminal-active rectangle equals the literal terminal incidence. When the test fails no parent-complete terminal-faithful pair-rectangular lift exists, and our canonical fallback retains the literal active hypergraph; when it passes, the exact excess parent mass is Δ_m. This yields a canonical hybrid repair and

$$\widetilde{P}_m = P_m + \Delta_m, \qquad \mathfrak{I}_X = \begin{cases} +\infty, & \exists m: \ P_m = 0 < \widetilde{P}_m, \\ \displaystyle\max_{m:P_m>0} \frac{\widetilde{P}_m}{P_m}, & \text{otherwise}, \end{cases}$$

for a nonzero literal parent. The literal-active-hypergraph baseline has exactly zero parent inflation. A logarithmic number of block labels does not control $\mathfrak{I}_X$: arbitrary polynomial inflation can occur in one block.

True terminal activation factors, on positive rows, as

$$\frac{T_m}{\widetilde{P}_m} = \frac{H_m}{P_m} \frac{T_m}{H_m} \frac{P_m}{\widetilde{P}_m}.$$

We give zero-safe conventions, sharp uniform row minima, exact distinguished weighted averages, and the optimal transfer $\tau_{\mathrm{rep}} \geq \tau_{\mathrm{hyp}}/\mathfrak{I}_X$. For a coefficient vector satisfying a cap and an occupied-capacity lower bound, the threshold data $g_m = T_m/\widetilde{P}_m \geq \eta$ on a repaired-parent fraction δ_η give the sharp certificate

$$\frac{T(z)}{\widetilde{P}(z)} \geq \eta \left(1 - \frac{1-\delta_\eta}{\kappa_X}\right)_+.$$

The certified right-hand side is positive exactly when $\kappa_X > 1 - \delta_\eta$.

Finally, we state a denominator-compatible chain and charge parent inflation once inside the effective terminal loss before applying the strict $1/400$ endpoint test. No favorable bound for T_m/P_m, δ_η, κ_X, physical label erasure, or a signed fixed-shift kernel is proved. The result is an L0/L1 activation-interface theorem, not a parity estimate, a prime-pair lower bound, or a twin-prime result.

Keywords: fixed shift; full incidence; terminal activation; parent inflation; multiscale repair; capacity saturation.

Convention. Every physical specialization uses one fixed nonzero shift h_0, literal coefficient-free support, and raw counting-measure atomic norms. The symbol h_0 is not specialized to 2. No support, activation, or capacity identity is promoted to a prime-pair or parity statement.

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1 Introduction

The fixed-shift program now has two distinct notions of "completion." TPC–63 completes the logical provenance interface: a pair relation may be rectangularized only after its literal full incidence has passed the appropriate parent, terminal, or grouped-output test [6]. It also proves that retaining dyadic cofactor scales as orthogonal labels costs only $X^{o(1)}$ at the raw-energy level. Neither statement says that the chosen completed space has the same parent denominator as the coefficientwise physical field of TPC–47 [1].

That denominator matters. Suppose a rectangle adds coordinates which are terminal-null but carry nonzero pre-terminal coefficient. Its terminal numerator is unchanged, while its parent form is larger. Parseval on the enlarged space is exact but answers the wrong comparison unless the extra parent mass is recorded. Conversely, replacing the parent by the terminal-active incidence deletes the very alignment defect which the activation theorem is meant to estimate. The present paper isolates this missing arrow before any represented-Gram or signed-cancellation argument is attempted.

For each dyadic block k, let $\mathcal{U}_{X,k}$ be the literal full incidence and let

$$\mathcal{U}_{X,k}^{\mathrm{pre},+} = \{u : c_u \neq 0\}, \qquad \mathcal{U}_{X,h,k}^{+} = \{u : t_h(u)c_u \neq 0\}.$$

The literal row masses are

$$P_m = \sum_k \sum_{\substack{u \in \mathcal{U}_{X,k}^{\mathrm{pre},+} \\ m(u)=m}} |c_u|^2, \tag{1}$$

$$H_m = \sum_k \sum_{\substack{u \in \mathcal{U}_{X,h,k}^{+} \\ m(u)=m}} |c_u|^2, \qquad T_m = \sum_k \sum_{\substack{u \in \mathcal{U}_{X,h,k}^{+} \\ m(u)=m}} |t_h(u)c_u|^2. \tag{2}$$

All three are fixed before the free row coefficient is chosen. In the physical specialization, t_h contains one inherited cofactor sign and the literal smooth terminal multiplier. It is not normalized to a probability.

The first result is an exact existence theorem. Put

$$I_{\mathrm{pre},k} := \mathrm{pr}_{m,r}\, \mathcal{U}_{X,k}^{\mathrm{pre},+}.$$

A pair relation which retains every literal parent atom and introduces no terminal-active spurious cross exists exactly when

$$\mathrm{TRect}_{h,k}(I_{\mathrm{pre},k}) = \mathcal{U}_{X,h,k}^{+}. \tag{3}$$

When it exists, $I_{\mathrm{pre},k}$ is the unique least relation. Its parent is uninflated exactly when its pre-terminal rectangle is already the literal active incidence. Blocks which fail (3) admit no parent-complete faithful pair-rectangular lift. Our canonical fallback retains their literal active incidence, giving a pair-good/hypergraph hybrid.

The second result measures the cost of every populated repair. With Δ_m the structural, z-independent squared mass of new terminal-null parent coordinates,

$$\widetilde{P}_m = P_m + \Delta_m. \tag{4}$$

The optimal full-row comparison is the largest row inflation; a new-only row makes the comparison infinite. The literal-active-hypergraph realization has $\Delta_m = 0$ identically. This distinguishes parent inflation from the Mellin-frame conditioning of TPC–51, the capacity saturation of TPC–54, and the physical label-erasure loss of TPC–62 [2, 3, 5].

The third result gives the correct activation quantity. On rows with all denominators positive,

$$\frac{T_m}{\widetilde{P}_m} = \underbrace{\frac{H_m}{P_m}}_{\text{terminal-support alignment}} \underbrace{\frac{T_m}{H_m}}_{\text{literal multiplier intensity}} \underbrace{\frac{P_m}{\widetilde{P}_m}}_{\text{inverse inflation}} . \tag{5}$$

The paper extends this identity through all zero rows, proves the sharp uniform and distinguished minimax laws, and shows that the loss $\mathfrak{I}_X^{-1}$ in passing from the literal hypergraph parent to the repaired parent cannot be improved from representation data alone.

There remains an honest distinguished alternative inherited from the sharp energy-selection interface of TPC–54. If good rows $g_m = T_m/\widetilde{P}_m \geq \eta$ occupy a structural repaired-parent fraction δ_η, while a coefficient vector occupies at least a fraction κ_X of the available capped energy, then

$$\frac{T(z)}{\widetilde{P}(z)} \geq \eta \left(1 - \frac{1-\delta_\eta}{\kappa_X}\right)_+ . \tag{6}$$

We prove that this bound and the threshold $\kappa_X > 1 - \delta_\eta$ are sharp from the stated data. The theorem does not assert that either input is favorable on the physical vector.

The compatible endpoint chain uses the same carrier, row vector, raw norm, and uniform or distinguished scope throughout TPC–47, TPC–51, TPC–54, TPC–59, TPC–62, and TPC–63 [1–6]. If a literal terminal estimate is first proved relative to P, parent inflation is charged once in its effective loss. If a direct estimate is proved relative to $\widetilde{P}$, it is not charged again. Equality at the inherited endpoint $1/400$ is a stop, not a success.

Main results. The paper proves the following finite-scale statements.

(i) Exact literal parent, terminal-support, and shaped terminal forms, with block disintegration and raw atomic normalization.

(ii) A necessary and sufficient parent-complete/terminal-faithful pair-lift criterion, its unique least relation, and the exact zero-inflation test.

(iii) A canonical hybrid repair and an exact literal-active-hypergraph zero-inflation baseline.

(iv) Exact parent-inflation identities, the optimal domination constant, and sharp arbitrary-inflation and zero-row countermodels.

(v) Zero-safe three-factor activation, uniform row minima, distinguished weighted averages, and sharp inflation-corrected transfer.

(vi) The sharp threshold-capacity certificate (6), including all endpoint extremizers.

(vii) A denominator-compatible five-arrow chain, a no-double-charge exponent ledger, and the strict $1/400$ stop rule.

The incidence and minimax theorems are L0. Their use with one literal fixed-shift coefficient field is L1 and remains conditional on an arithmetic terminal-incidence estimate. No L2 cancellation, parity, prime-pair, or twin-prime conclusion is claimed.

2 The literal multiscale parent and its terminal field

The multiscale construction must begin before pair projection. This section fixes the literal finite incidence, its raw parent form, and the terminal form to which every later repair will be compared. In particular, neither a rectangular completion nor the terminal-active support is used to define the parent.

2.1 Finite arithmetic blocks and full incidences

Fix $X \geq 2$ and one integer

$$h \in \mathbb{Z} \setminus \{0\}. \tag{7}$$

The physical specialization will take $h = h_0$, but no value of h_0 is selected here. Let $K_X \subset \mathbb{Z}$ be a finite set of block labels. For $k \in K_X$, let $\mathcal{R}_{X,k} \subset \mathbb{N}$ be finite, with the $\mathcal{R}_{X,k}$ pairwise disjoint, and let

$$\Omega_{X,k} \subseteq \mathcal{M}_X \times \mathcal{R}_{X,k} \times \mathcal{J}_X \times \Gamma_X \tag{8}$$

be a finite coefficient-independent ambient incidence. Here $\mathcal{M}_X, \mathcal{J}_X \subset \mathbb{N}$ and Γ_X is a finite auxiliary label set. A point is written

$$u = (m, r, j, \gamma), \qquad m(u) = m, \quad r(u) = r, \quad j(u) = j, \quad \gamma(u) = \gamma.$$

Allowing $\Omega_{X,k}$ to be a subset of the displayed product permits fixed residue, side, orbit, and support restrictions. They are part of the ambient arithmetic data and are fixed before a row coefficient is chosen.

For every k, let

$$\mathcal{U}_{X,k} \subseteq \Omega_{X,k} \tag{9}$$

be the literal full pre-terminal incidence on that block. The word *full* means that all four labels are retained; it does not mean that $\mathcal{U}_{X,k}$ is a Cartesian product. Attach a structural coefficient

$$c_u \in \mathbb{C}, \qquad u \in \Omega_{X,k}, \tag{10}$$

containing every coefficient-free source, mask, opposite-row, residue, and orbit factor preceding terminalization. Both $\mathcal{U}_{X,k}$ and c_u are fixed independently of

$$z = (z_m)_{m \in \mathcal{M}_X} \in Z_X := \mathbb{C}^{\mathcal{M}_X}.$$

Thus a zero of z_m never deletes an atom from the structural support.

2.2 The literal terminal multiplier

For $u = (m, r, j, \gamma)$, put

$$n_h(u) := mj + h. \tag{11}$$

If $n_h(u) > 0$ and $r \mid n_h(u)$, define

$$p_h(u) := \frac{n_h(u)}{r} \in \mathbb{N}. \tag{12}$$

Otherwise $p_h(u)$ is left undefined and the terminal indicator below is zero. Let $\mathbf{1}_{\mathrm{term},h}(u) \in \{0, 1\}$ be the conjunction of the complete declared terminal conditions. In the fixed-shift arithmetic specialization these include

$$n_h(u) > 0, \qquad r \mid n_h(u), \qquad p_h(u) \in \mathbb{P}, \tag{13}$$

the declared prime and cofactor windows, and every inherited squarefree, coprimality, literal-mask, and terminal-profile support condition. No test is allowed to depend on z.

Let $\Lambda_X : \mathbb{N} \to \mathbb{C}$ be the literal shaped terminal weight and define

$$t_h(u) := \mu(r(u))\Lambda_X(n_h(u))\mathbf{1}_{\mathrm{term},h}(u). \tag{14}$$

The factor $\mu(r)$ occurs once. If the physical packet uses an equivalent calibrated multiplier, that multiplier is inserted in (14) without changing any argument below.

Definition 2.1 (Literal active incidences). For $k \in K_X$, define

$$\mathcal{U}_{X,k}^{\mathrm{pre},+} := \{u \in \mathcal{U}_{X,k} : c_u \neq 0\}, \tag{15}$$

$$\mathcal{U}_{X,h,k}^{+} := \{u \in \mathcal{U}_{X,k} : t_h(u)c_u \neq 0\}. \tag{16}$$

Both sets are coefficient-independent and

$$\mathcal{U}_{X,h,k}^{+} \subseteq \mathcal{U}_{X,k}^{\mathrm{pre},+}. \tag{17}$$

The inclusion may be strict. Replacing the first set by the second in an upstream parent form would discard terminal-null parent mass and make activation circular.

2.3 Raw parent, support-aligned, and terminal forms

For $m \in \mathcal{M}_X$ and $k \in K_X$, define

$$P_{k,m} := \sum_{\substack{u \in \mathcal{U}_{X,k}^{\mathrm{pre},+} \\ m(u)=m}} |c_u|^2, \tag{18}$$

$$H_{k,m} := \sum_{\substack{u \in \mathcal{U}_{X,h,k}^{+} \\ m(u)=m}} |c_u|^2, \tag{19}$$

$$T_{k,m} := \sum_{\substack{u \in \mathcal{U}_{X,h,k}^{+} \\ m(u)=m}} |t_h(u)c_u|^2. \tag{20}$$

An empty sum is zero. Aggregate the blocks by

$$P_m := \sum_{k \in K_X} P_{k,m}, \qquad H_m := \sum_{k \in K_X} H_{k,m}, \qquad T_m := \sum_{k \in K_X} T_{k,m}. \tag{21}$$

Then

$$0 \le H_m \le P_m, \qquad P_m = 0 \Longrightarrow H_m = T_m = 0. \tag{22}$$

There is no universal order between T_m and H_m, because the literal multiplier $|\Lambda_X|^2$ remains in T_m.

The corresponding quadratic forms on the original row-coefficient space are

$$\mathcal{P}_X(z) := \sum_{m \in \mathcal{M}_X} P_m |z_m|^2, \tag{23}$$

$$\mathcal{H}_{X,h}(z) := \sum_{m \in \mathcal{M}_X} H_m |z_m|^2, \tag{24}$$

$$\mathcal{T}_{X,h}(z) := \sum_{m \in \mathcal{M}_X} T_m |z_m|^2. \tag{25}$$

These are raw counting-measure forms. Repeated cofactors, orbit points, or auxiliary labels remain distinct orthogonal atoms. In particular, there is no division by a row multiplicity, cofactor multiplicity, or terminal fiber size.

Equivalently, on the labeled spaces

$$\mathcal{H}_X^{\mathrm{pre}} := \bigoplus_{k\in K_X}^{\perp} \ell^2(\mathcal{U}_{X,k}^{\mathrm{pre},+}), \qquad \mathcal{H}_{X,h}^{\mathrm{term}} := \bigoplus_{k\in K_X}^{\perp} \ell^2(\mathcal{U}_{X,h,k}^{+}), \tag{26}$$

Write $\mathcal{U}^{\mathrm{pre},+} := (\mathcal{U}_{X,k}^{\mathrm{pre},+})_{k\in K_X}$ for the literal parent family. For any family $\mathcal{V} = (\mathcal{V}_{X,k})_{k\in K_X}$ satisfying

$$\mathcal{U}_{X,h,k}^{+} \subseteq \mathcal{V}_{X,k} \subseteq \Omega_{X,k}, \qquad k \in K_X,$$

define the typed terminal operator

$$D_{X,h}^{\mathcal{V}} : \bigoplus_{k\in K_X}^{\perp} \ell^2(\mathcal{V}_{X,k}) \longrightarrow \mathcal{H}_{X,h}^{\mathrm{term}}, \qquad (D_{X,h}^{\mathcal{V}} a)(k,u) := t_h(u)a(k,u) \quad (u \in \mathcal{U}_{X,h,k}^{+}).$$

Thus the operator domain changes with the populated parent incidence, while the terminal codomain remains literal. The literal analysis and terminal maps are

$$(C_X^{\mathrm{lit}} z)(k,u) := c_u z_{m(u)}, \tag{27}$$

$$(D_{X,h}^{\mathcal{U}^{\mathrm{pre},+}} C_X^{\mathrm{lit}} z)(k,u) := t_h(u) c_u z_{m(u)}. \tag{28}$$

Thus

$$\|C_X^{\mathrm{lit}} z\|_2^2 = \mathcal{P}_X(z), \qquad \|D_{X,h}^{\mathcal{U}^{\mathrm{pre},+}} C_X^{\mathrm{lit}} z\|_2^2 = \mathcal{T}_{X,h}(z). \tag{29}$$

Proposition 2.2 (Exact block disintegration)**.** *For every $z \in Z_X$,*

$$\mathcal{P}_X(z) = \sum_{k\in K_X} \sum_m P_{k,m} |z_m|^2, \tag{30}$$

$$\mathcal{T}_{X,h}(z) = \sum_{k\in K_X} \sum_m T_{k,m} |z_m|^2. \tag{31}$$

These are exact identities with constant one, independently of $|K_X|$.

Proof. The external block labels make the summands orthogonal, and the cofactor blocks are disjoint. Expanding the two raw squared norms in (29) and regrouping first by k and then by m gives the displayed identities. □

2.4 The activation comparison is not yet an estimate

The literal activation problem is the comparison of $\mathcal{T}_{X,h}$ with $\mathcal{P}_X$, in one declared uniform or distinguished coefficient scope. If $\mathcal{P}_X \equiv 0$, there is no parent-visible vector and no activation constant or exponent is assigned. If $\mathcal{P}_X \not\equiv 0$, the definitions above do not assert that every active parent row has $T_m > 0$, nor that any positive ratio has a fixed-power lower bound.

Normalizing $\mathcal{T}_{X,h}$ by $\mathcal{H}_{X,h}$, or declaring $\mathcal{U}_{X,h,k}^{+}$ to be the parent incidence, would delete the parent-to-terminal-support alignment loss. Such a terminal-active parent cannot be substituted for $\mathcal{P}_X$ in a chain whose upstream reference is the literal pre-terminal form.

3 Parent-complete terminal-faithful pair lifts

A pair relation can be useful only if its rectangular lift represents the same terminal field without deleting the parent against which activation is measured. This section gives the exact existence test, the unique least relation, and the resulting parent inflation.

3.1 Block rectangles and two active closures

Fix $k \in K_X$. For a relation

$$I \subseteq \mathcal{M}_X \times \mathcal{R}_{X,k},$$

define its block rectangle inside the declared ambient incidence by

$$\mathrm{Rect}_k(I) := \{u \in \Omega_{X,k} : (m(u), r(u)) \in I\}. \tag{32}$$

Its pre-terminal and terminal-active closures are

$$\mathrm{PRect}_k(I) := \{u \in \mathrm{Rect}_k(I) : c_u \neq 0\}, \tag{33}$$

$$\mathrm{TRect}_{h,k}(I) := \{u \in \mathrm{Rect}_k(I) : t_h(u)c_u \neq 0\}. \tag{34}$$

Both are monotone in I.

Definition 3.1 (Parent-complete and terminal-atomic faithful). A pair relation I on block k is

(i) *parent-complete* if

$$\mathcal{U}^{\mathrm{pre},+}_{X,k} \subseteq \mathrm{PRect}_k(I); \tag{35}$$

(ii) *terminal-atomic faithful* if

$$\mathrm{TRect}_{h,k}(I) = \mathcal{U}^{+}_{X,h,k}. \tag{36}$$

Parent completeness retains every nonzero literal pre-terminal coordinate. Terminal-atomic faithfulness demands equality before any coherent grouping; terminal-active spurious crosses are therefore forbidden even when a later grouping might cancel them.

Put

$$I_{\mathrm{pre},k} := \mathrm{pr}_{m,r}\,\mathcal{U}^{\mathrm{pre},+}_{X,k}. \tag{37}$$

By construction,

$$I \text{ is parent-complete} \iff I_{\mathrm{pre},k} \subseteq I. \tag{38}$$

3.2 The exact existence and minimality theorem

Theorem 3.2 (Parent-complete faithful-lift criterion)**.** *There exists a pair relation on block k which is simultaneously parent-complete and terminal-atomic faithful if and only if*

$$\boxed{\mathrm{TRect}_{h,k}(I_{\mathrm{pre},k}) = \mathcal{U}^{+}_{X,h,k}.} \tag{39}$$

When (39) holds, $I_{\mathrm{pre},k}$ is the unique least such relation. More precisely, I is parent-complete and terminal-atomic faithful if and only if

$$I_{\mathrm{pre},k} \subseteq I \quad \text{and} \quad \mathrm{TRect}_{h,k}(I \setminus I_{\mathrm{pre},k}) = \varnothing. \tag{40}$$

Proof. Every point of $\mathcal{U}^{+}_{X,h,k}$ belongs to $\mathcal{U}^{\mathrm{pre},+}_{X,k}$, so its pair lies in $I_{\mathrm{pre},k}$. Hence

$$\mathcal{U}^{+}_{X,h,k} \subseteq \mathrm{TRect}_{h,k}(I_{\mathrm{pre},k}). \tag{41}$$

If I is parent-complete, then $I_{\mathrm{pre},k} \subseteq I$ by (38). If it is also terminal-atomic faithful, monotonicity gives

$$\mathcal{U}^{+}_{X,h,k} \subseteq \mathrm{TRect}_{h,k}(I_{\mathrm{pre},k}) \subseteq \mathrm{TRect}_{h,k}(I) = \mathcal{U}^{+}_{X,h,k},$$

which proves necessity of (39). Conversely, that equality makes $I_{\mathrm{pre},k}$ both parent-complete and faithful. Every parent-complete relation contains it, proving unique leastness.

Assume the criterion. Adding pairs outside $I_{\mathrm{pre},k}$ preserves terminal equality exactly when those pairs create no terminal-active point, which is the second condition in (40). The first condition is exactly parent completeness. □

If (39) fails, every pair relation which retains all literal parent atoms creates a terminal-active spurious cross. No enlargement of the relation can remove that cross. Hence no parent-complete, terminal-atomic faithful pair-rectangular lift exists. The canonical repair used below then falls back to the literal active hypergraph. This statement excludes pair rectangles only; it makes no minimality claim among arbitrary nonlinear, quotient, or grouped representations.

3.3 Exact parent inflation

For a parent-complete relation I, define its populated rectangular row weight by

$$\widetilde{P}_{k,m}(I) := \sum_{\substack{u\in \mathrm{PRect}_k(I)\\ m(u)=m}} |c_u|^2 \tag{42}$$

and its added row mass by

$$\Delta_{k,m}(I) := \sum_{\substack{u\in \mathrm{PRect}_k(I)\setminus \mathcal{U}^{\mathrm{pre},+}_{X,k}\\ m(u)=m}} |c_u|^2. \tag{43}$$

Because parent completeness gives inclusion of the literal active incidence, the union is disjoint and

$$\boxed{\widetilde{P}_{k,m}(I) = P_{k,m} + \Delta_{k,m}(I).} \tag{44}$$

Likewise define

$$\widetilde{T}_{k,m}(I) := \sum_{\substack{u\in \mathrm{TRect}_{h,k}(I)\\ m(u)=m}} |t_h(u)c_u|^2. \tag{45}$$

If I is terminal-atomic faithful, then

$$\boxed{\widetilde{T}_{k,m}(I) = T_{k,m}} \tag{46}$$

for every row. Thus a terminal-faithful rectangle can change the denominator while leaving the terminal numerator exactly unchanged.

Theorem 3.3 (Minimal parent inflation and its zero test)**.** *Assume* (39)*. Then, for every parent-complete terminal-atomic faithful relation I,*

$$\Delta_{k,m}(I_{\mathrm{pre},k}) \le \Delta_{k,m}(I) \qquad (m \in \mathcal{M}_X). \tag{47}$$

The canonical pair lift has zero parent inflation on every row if and only if the literal pre-terminal active incidence is rectangular over its pair projection:

$$\boxed{\Delta_{k,m}(I_{\mathrm{pre},k}) = 0 \text{ for every } m \iff \mathrm{PRect}_k(I_{\mathrm{pre},k}) = \mathcal{U}^{\mathrm{pre},+}_{X,k}.} \tag{48}$$

If the terminal criterion holds but (48) *fails, then some row has strictly positive added parent mass, and every added nonzero coordinate is terminal-null.*

Proof. Every admissible I contains $I_{\mathrm{pre},k}$. Monotonicity of PRect_k, followed by termwise nonnegativity in (43), proves (47). The inclusion

$$\mathcal{U}^{\mathrm{pre},+}_{X,k} \subseteq \mathrm{PRect}_k(I_{\mathrm{pre},k})$$

always holds. Therefore all rowwise added sums vanish exactly when the two finite active sets are equal, proving (48). If they are unequal, a spurious point has $c_u \neq 0$ and contributes positively to its row. Terminal faithfulness forces $t_h(u)c_u = 0$ at every such point. ∎

For $P_{k,m} > 0$, define the block inflation

$$\iota_{k,m}(I) := \frac{\widetilde{P}_{k,m}(I)}{P_{k,m}} = 1 + \frac{\Delta_{k,m}(I)}{P_{k,m}}. \tag{49}$$

If $P_{k,m} = \widetilde{P}_{k,m}(I) = 0$, put $\iota_{k,m}(I) = 1$. If $P_{k,m} = 0 < \widetilde{P}_{k,m}(I)$, put $\iota_{k,m}(I) = +\infty$. The last case cannot occur for the canonical relation $I_{\mathrm{pre},k}$, but it can occur after arbitrary extra rows are inserted.

The identities (44) and (46) show exactly which denominator a later activation theorem must use. They do not by themselves give a positive activation constant.

Remark 3.4 (Terminal-only least relations do not preserve the parent). The smaller relation $\mathrm{pr}_{m,r}\,\mathcal{U}^{+}_{X,h,k}$ may represent the terminal atomic field while omitting pairs which carry nonzero pre-terminal mass. A favorable ratio obtained after that deletion is not a lower bound relative to $P_{k,m}$. Parent completeness is precisely the condition preventing this substitution.

The exact size of the resulting activation dilution, new-only row failures, and sharp finite countermodels are separate from the existence theorem. They must be evaluated from the populated repaired parent, not from the number of pair or block labels.

4 Canonical hybrid repair and the activation-inflation law

The blockwise criterion permits a canonical repair without forcing a pair-only model on blocks where pair projection is unfaithful. Pair-good blocks use their unique least parent-complete rectangle; pair-bad blocks retain their literal active incidence. All block labels remain orthogonal until a separately declared physical erasure.

4.1 Pair-good and hypergraph blocks

Define

$$K_X^{\mathrm{pair}} := \left\{ k \in K_X : \mathrm{TRect}_{h,k}(I_{\mathrm{pre},k}) = \mathcal{U}^{+}_{X,h,k} \right\}, \tag{50}$$

$$K_X^{\mathrm{hyp}} := K_X \setminus K_X^{\mathrm{pair}}. \tag{51}$$

For each block, define the canonical repaired parent incidence

$$\mathcal{V}^{\mathrm{hyb}}_{X,k} := \begin{cases} \mathrm{PRect}_k(I_{\mathrm{pre},k}), & k \in K_X^{\mathrm{pair}}, \\ \mathcal{U}^{\mathrm{pre},+}_{X,k}, & k \in K_X^{\mathrm{hyp}}. \end{cases} \tag{52}$$

Write $\mathcal{V}^{\mathrm{hyb}} := (\mathcal{V}^{\mathrm{hyb}}_{X,k})_{k\in K_X}$ for this populated parent family. The external label k is retained even if two underlying ambient coordinates would otherwise be written with the same tuple.

The literal hypergraph baseline and the hybrid parent spaces are

$$\mathcal{H}_X^{\mathrm{hyp}} := \bigoplus_{k\in K_X}^{\perp} \ell^2(\mathcal{U}^{\mathrm{pre},+}_{X,k}), \tag{53}$$

$$\mathcal{H}_X^{\mathrm{hyb}} := \bigoplus_{k\in K_X}^{\perp} \ell^2(\mathcal{V}^{\mathrm{hyb}}_{X,k}). \tag{54}$$

Their populated analyses use the same literal field:

$$(C_X^{\mathrm{hyp}} z)(k,u) := c_u z_{m(u)}, \tag{55}$$

$$(C_X^{\mathrm{hyb}} z)(k,u) := c_u z_{m(u)}. \tag{56}$$

The formulas have different coordinate domains. In particular, C_X^{hyb} populates the terminal-null spurious crosses on pair-good blocks. Merely embedding $\mathcal{H}_X^{\mathrm{hyp}}$ into a larger ambient space by zero extension would not do so and would have no parent inflation.

4.2 Exact hybrid identities

Put

$$\Delta_m^{\mathrm{hyb}} := \sum_{k \in K_X^{\mathrm{pair}}} \Delta_{k,m}(I_{\mathrm{pre},k}) \tag{57}$$

and

$$\widetilde{P}_m := P_m + \Delta_m^{\mathrm{hyb}}. \tag{58}$$

The hybrid parent form is

$$\widetilde{\mathcal{P}}_X(z) := \|C_X^{\mathrm{hyb}} z\|_2^2 = \sum_m \widetilde{P}_m |z_m|^2. \tag{59}$$

Theorem 4.1 (Canonical hybrid repair)**.** *For every $z \in Z_X$,*

$$\widetilde{\mathcal{P}}_X(z) = \mathcal{P}_X(z) + \sum_m \Delta_m^{\mathrm{hyb}} |z_m|^2, \tag{60}$$

$$\|D_{X,h}^{\mathcal{V}^{\mathrm{hyb}}} C_X^{\mathrm{hyb}} z\|_2^2 = \mathcal{T}_{X,h}(z). \tag{61}$$

Thus the hybrid repair is parent-complete and terminal-atomic faithful on every block. Among all repairs which use a parent-complete terminal-atomic faithful pair relation on every block in K_X^{pair} and retain the literal active incidence on every block in K_X^{hyp}, it minimizes the parent weight row by row.

The literal-active-hypergraph choice has the stronger exact identities

$$\boxed{\|C_X^{\mathrm{hyp}} z\|_2^2 = \mathcal{P}_X(z), \qquad \|D_{X,h}^{\mathcal{U}^{\mathrm{pre},+}} C_X^{\mathrm{hyp}} z\|_2^2 = \mathcal{T}_{X,h}(z).} \tag{62}$$

Hence the literal active hypergraph baseline has exactly zero parent inflation, not merely zero fixed-power inflation.

Proof. On a pair-good block, (44) and (46) apply to $I_{\mathrm{pre},k}$. On a hypergraph block both forms are the literal ones by definition. Orthogonality of the external block labels allows the identities to be summed over k, proving (60) and (61). Rowwise minimality follows from (47) on every pair-good block. For the literal-active-hypergraph choice, every block is the original literal active coordinate set, so exact block disintegration gives (62). □

No theorem requires using a pair rectangle on a pair-good block. Within populated incidence repairs retaining every literal parent atom, the literal active hypergraph is the parent-minimal baseline. The hybrid construction records the least cost incurred when a downstream argument explicitly requires pair coordinates on the blocks where such a representation is faithful.

4.3 Block count is not an inflation estimate

Suppose the cofactor blocks lie in a fixed polynomial range. Then the dyadic choice has

$$|K_X| = O(\log X) = X^{o(1)}. \tag{63}$$

This controls only the number of external labels. It does not control Δ_m^{hyb}, or any ratio of repaired to literal parent mass. Parent inflation of arbitrary polynomial size can already occur inside a single pair-good block.

For clarity, put

$$\widetilde{P}_{k,m}^{\mathrm{hyb}} := \begin{cases} P_{k,m} + \Delta_{k,m}(I_{\mathrm{pre},k}), & k \in K_X^{\mathrm{pair}}, \\ P_{k,m}, & k \in K_X^{\mathrm{hyp}}. \end{cases} \tag{64}$$

If a separate estimate proves, for one common finite I_X,

$$\widetilde{P}^{\mathrm{hyb}}_{k,m} \le I_X P_{k,m} \tag{65}$$

on every block and active row, then summing gives

$$\widetilde{P}_m = \sum_k \widetilde{P}^{\mathrm{hyb}}_{k,m} \le I_X \sum_k P_{k,m} = I_X P_m. \tag{66}$$

Indeed, $P_{k,m} = 0$ implies $\widetilde{P}^{\mathrm{hyb}}_{k,m} = 0$: no pair with first coordinate m then occurs in $I_{\mathrm{pre},k}$, and a hypergraph block keeps the same empty active row. Thus (65) also holds trivially on every inactive row, justifying the sum over all k. No additional factor $|K_X|$ is needed in this comparison. Conversely, (63) supplies no value of I_X. The label count and the parent-inflation bound are logically independent data.

Erasing the external label k, coherently grouping residual labels, or mapping the terminal vector to a native physical output is a later map. Neither (63) nor (62) supplies a lower bound through that map. Its kernel and range-restricted Gram belong to the reassembly gate, not to parent inflation.

Remark 4.2 (Claim boundary). All results in this section are exact finite-incidence statements. Their use with the literal multiplier (14), one fixed nonzero shift, and the raw physical atom norm is an arithmetic-interface specialization. No estimate here proves that $T_m > 0$, gives a polynomial terminal-activation lower bound, specializes h to 2, estimates a signed parity kernel, or implies a prime-pair statement.

5 Exact parent-inflation laws

The terminal-faithfulness test does not by itself preserve the parent denominator. A pair rectangle may add coordinates which are terminal-null but carry nonzero pre-terminal mass. This section records that mass exactly. All incidences, coefficients, and repairs are fixed before the row coefficient vector is chosen.

5.1 Repaired parent data

Continue with the literal blocks, incidences, and row masses

$$\mathcal{U}^{\mathrm{pre},+}_{X,k}, \qquad \mathcal{U}^{+}_{X,h,k}, \qquad P_m,\ H_m,\ T_m$$

from section 2. In particular, P_m, H_m, T_m are not recomputed after a pair projection or terminal restriction.

Fix a populated parent enlargement $\widetilde{\mathcal{U}}^{\mathrm{pre},+}_{X,k}$ which preserves every literal parent atom and is terminal-atomic faithful. Thus

$$\mathcal{U}^{\mathrm{pre},+}_{X,k} \subseteq \widetilde{\mathcal{U}}^{\mathrm{pre},+}_{X,k} \subseteq \{u \in \Omega_{X,k} : c_u \neq 0\}, \qquad \{u \in \widetilde{\mathcal{U}}^{\mathrm{pre},+}_{X,k} : t_h(u)c_u \neq 0\} = \mathcal{U}^{+}_{X,h,k}. \tag{67}$$

The added-parent incidence

$$\mathcal{N}_{X,k} := \widetilde{\mathcal{U}}^{\mathrm{pre},+}_{X,k} \setminus \mathcal{U}^{\mathrm{pre},+}_{X,k} \tag{68}$$

therefore satisfies

$$t_h(u)c_u = 0 \qquad (u \in \mathcal{N}_{X,k}). \tag{69}$$

For a pair lift, the first condition is parent completeness in the sense of definition 3.1. For the canonical hybrid repair, take $\widetilde{\mathcal{U}}^{\mathrm{pre},+}_{X,k} = \mathcal{V}^{\mathrm{hyb}}_{X,k}$: this is the least admissible parent-complete rectangle on a pair-good block and the literal active incidence on a pair-bad block. The results below use only (67), so they also apply to any larger populated terminal-faithful repair.

For every source row m, define only the genuinely new quantity

$$\Delta_m := \sum_{k\in K_X} \sum_{\substack{u\in\mathcal{N}_{X,k}\\ m(u)=m}} |c_u|^2. \tag{70}$$

The populated repaired parent mass therefore obeys

$$\widetilde{P}_m = P_m + \Delta_m. \tag{71}$$

In the canonical hybrid specialization, $\Delta_m = \Delta_m^{\mathrm{hyb}}$, and (71) is exactly the previously defined hybrid row parent (58); no second hybrid parent is being introduced. All four quantities are nonnegative, and

$$0 \le H_m \le P_m \le \widetilde{P}_m. \tag{72}$$

Terminal faithfulness means that the repair changes neither H_m nor T_m. It changes only the parent against which terminal activation is measured.

For $z = (z_m)_{m\in\mathcal{M}_X} \in Z_X$, set

$$P(z) := \mathcal{P}_X(z) = \sum_m P_m|z_m|^2, \qquad \Delta(z) := \sum_m \Delta_m|z_m|^2, \tag{73}$$

$$\widetilde{P}(z) := \sum_m \widetilde{P}_m|z_m|^2, \qquad T(z) := \mathcal{T}_{X,h}(z) = \sum_m T_m|z_m|^2. \tag{74}$$

We assume throughout that P is not identically zero. Rows on which $P_m = \widetilde{P}_m = 0$ are harmless zero rows.

5.2 The exact inflation constant

The row inflation ratio is an extended nonnegative number. We use the convention

$$\iota_m := \begin{cases} \dfrac{\widetilde{P}_m}{P_m} = 1 + \dfrac{\Delta_m}{P_m}, & P_m > 0, \\ 1, & P_m = \widetilde{P}_m = 0, \\ +\infty, & P_m = 0 < \widetilde{P}_m. \end{cases} \tag{75}$$

A row in the last case is called a *new-only parent row*. Define

$$\mathfrak{I}_X := \sup_{\substack{z\in Z_X\\ P(z)>0}} \frac{\widetilde{P}(z)}{P(z)} \in [1, +\infty]. \tag{76}$$

The supremum is taken on the full row-coefficient space. A range-restricted or distinguished coefficient class has its own constant and must not be substituted for $\mathfrak{I}_X$.

Theorem 5.1 (Exact parent inflation and sharp domination)**.** *For every coefficient vector,*

$$\boxed{\widetilde{P}(z) = P(z) + \Delta(z).} \tag{77}$$

Moreover,

$$\boxed{\mathfrak{I}_X = \max_{m\in\mathcal{M}_X} \iota_m} \tag{78}$$

with the conventions in (75)*. Consequently, the following are equivalent:*

(i) *there is a finite C such that $\widetilde{P}(z) \le CP(z)$ for every z;*

(ii) *the repair has no new-only parent row;*

(iii) $\mathfrak{I}_X < \infty$.

When these conditions hold, the least possible domination constant is $\mathfrak{I}_X$*:*

$$\widetilde{P}(z) \le \mathfrak{I}_X P(z) \qquad (z \in Z_X). \tag{79}$$

For the separate literal-active-hypergraph baseline, which makes no pair-rectangularization claim, $\Delta_m = 0$, $\widetilde{P}_m = P_m$, *and* $\mathfrak{I}_X = 1$ *exactly, consistently with* (62).

Proof. Equation (77) follows by summing (71) against $|z_m|^2$. First suppose there is no new-only row. Whenever $P(z) > 0$,

$$\frac{\widetilde{P}(z)}{P(z)} = \sum_{m:P_m>0} \frac{P_m|z_m|^2}{P(z)} \iota_m. \tag{80}$$

This is a convex combination of the active row ratios. It is at most their maximum, and a row basis vector attaining the maximum gives equality.

Now suppose $P_{m_1} = 0 < \widetilde{P}_{m_1}$. Choose a row m_0 with $P_{m_0} > 0$ and put

$$z^{(\varepsilon)} := e_{m_1} + \varepsilon e_{m_0}.$$

Then

$$P(z^{(\varepsilon)}) = \varepsilon^2 P_{m_0} > 0, \qquad \widetilde{P}(z^{(\varepsilon)}) \ge \widetilde{P}_{m_1},$$

so the quotient tends to $+\infty$. This proves (78) and all three equivalences. The hypergraph statement is the case $\widetilde{\mathcal{U}}^{\mathrm{pre},+}_{X,k} = \mathcal{U}^{\mathrm{pre},+}_{X,k}$ for every block. □

Proposition 5.2 (Sharpness of the inflation law)**.** *For every prescribed* $L \in [1, \infty)$*, there is a one-row, one-block terminal-faithful repair with*

$$P_m = 1, \qquad \Delta_m = L - 1, \qquad \widetilde{P}_m = L, \qquad \mathfrak{I}_X = L. \tag{81}$$

Thus neither the coefficient 1 *in* (77) *nor the constant* $\mathfrak{I}_X$ *in* (79) *can be improved under the stated hypotheses.*

Proof. Take one literal parent atom of square mass 1 and one added terminal-null parent atom of square mass $L - 1$, omitting the latter when $L = 1$. The repair is terminal-faithful because the added coordinate has zero terminal multiplier. All formulas are immediate. An explicit pair-rectangle realization, with arbitrarily large integral inflation, is given in example 8.1. □

The value $\mathfrak{I}_X$ is a parent-denominator constant. It is not the number of dyadic labels, a terminal occupancy proportion, an erasure norm, or a cancellation exponent. In particular, an $X^{o(1)}$ label count places no bound on $\mathfrak{I}_X$.

6 True terminal activation and its three exact factors

We now compare the shaped terminal energy with the repaired parent, not with a terminal-active parent chosen after the fact. The distinction produces three factors: literal support alignment, literal multiplier intensity, and inverse parent inflation. Only the first and third are proportions.

6.1 Zero-safe row factors

Continue with $P_m, H_m, T_m, \Delta_m, \widetilde{P}_m$ from section 5. The finite incidence definitions imply

$$H_m = 0 \implies T_m = 0, \qquad P_m = 0 \implies H_m = T_m = 0. \tag{82}$$

Define the support-alignment, multiplier-intensity, inverse-inflation, and repaired-activation factors by

$$\alpha_m := \begin{cases} H_m/P_m, & P_m > 0, \\ 0, & P_m = 0, \end{cases} \tag{83}$$

$$\vartheta_m := \begin{cases} T_m/H_m, & H_m > 0, \\ 0, & H_m = 0, \end{cases} \tag{84}$$

$$\rho_m := \begin{cases} P_m/\widetilde{P}_m, & \widetilde{P}_m > 0, \\ 1, & \widetilde{P}_m = 0, \end{cases} \tag{85}$$

$$g_m := \begin{cases} T_m/\widetilde{P}_m, & \widetilde{P}_m > 0, \\ 0, & \widetilde{P}_m = 0. \end{cases} \tag{86}$$

With $1/(+\infty) := 0$, the row conventions give $\rho_m = \iota_m^{-1}$ on every row. Thus $0 \le \alpha_m, \rho_m \le 1$, but there is no upper bound 1 on ϑ_m. Indeed, when $H_m > 0$,

$$\vartheta_m = \sum_{k \in K_X} \sum_{\substack{u \in \mathcal{U}^+_{X,h,k} \\ m(u) = m}} \frac{|c_u|^2}{H_m} |t_h(u)|^2. \tag{87}$$

It is a weighted average of squared literal multiplier magnitudes, not a probability of terminal survival. In particular, shaped logarithmic or profile weights can make it larger than one. On a row with a zero denominator, the displayed zero values are bookkeeping conventions chosen to make the universal product identity below valid. They are not conditional probabilities and do not assign a physical multiplier intensity to an empty support.

Theorem 6.1 (Exact three-factor activation law)**.** *On every row for which all three raw denominators are positive,*

$$\boxed{\frac{T_m}{\widetilde{P}_m} = \frac{H_m}{P_m} \frac{T_m}{H_m} \frac{P_m}{\widetilde{P}_m}}. \tag{88}$$

With the zero conventions (83)–(86), the universal statement is

$$\boxed{g_m = \alpha_m \vartheta_m \rho_m} \qquad (m \in \mathcal{M}_X). \tag{89}$$

More explicitly:

(i) *if $P_m, H_m, \widetilde{P}_m > 0$, then (88) holds by cancellation;*

(ii) *if $P_m > 0$ but $H_m = 0$, then $T_m = g_m = 0$;*

(iii) *if $P_m = 0 < \widetilde{P}_m$, then the row is new-only, $H_m = T_m = 0$, and $g_m = \rho_m = 0$;*

(iv) *if $\widetilde{P}_m = 0$, then all four row masses vanish, $g_m = 0$, and the harmless convention $\rho_m = 1$ agrees with the reciprocal of $\iota_m = 1$; the row contributes to no quadratic form.*

Proof. The regular identity is obtained by inserting H_m and P_m between the numerator and denominator. The remaining cases follow from (82) and $0 \le P_m \le \widetilde{P}_m$. □

The factorization separates three logically independent tasks. Parent-to-terminal support alignment controls α_m, the literal shaped coefficient controls ϑ_m, and the chosen repair controls ρ_m. Completing a relation can improve neither of the first two factors, while a terminal-faithful rectangle can strictly reduce the third.

6.2 Uniform and distinguished activation

Define the literal-hypergraph and repaired-parent uniform constants by

$$\tau_{\text{hyp}} := \inf_{\substack{z \in Z_X \\ P(z)>0}} \frac{T(z)}{P(z)}, \tag{90}$$

$$\tau_{\text{rep}} := \inf_{\substack{z \in Z_X \\ \widetilde{P}(z)>0}} \frac{T(z)}{\widetilde{P}(z)}. \tag{91}$$

These are full-row constants. They are not statements about one distinguished physical vector.

Theorem 6.2 (Sharp uniform row minima)**.** *On the full row-coefficient space,*

$$\boxed{\tau_{\text{hyp}} = \min_{m:P_m>0} \frac{T_m}{P_m} = \min_{m:P_m>0} \alpha_m \vartheta_m,} \tag{92}$$

$$\boxed{\tau_{\text{rep}} = \min_{m:\widetilde{P}_m>0} \frac{T_m}{\widetilde{P}_m} = \min_{m:\widetilde{P}_m>0} \alpha_m \vartheta_m \rho_m.} \tag{93}$$

In particular, one new-only parent row forces $\tau_{\text{rep}} = 0$, and one literal row with $P_m > 0 = T_m$ forces both uniform constants to vanish.

Proof. For $P(z) > 0$,

$$\frac{T(z)}{P(z)} = \sum_{m:P_m>0} \frac{P_m |z_m|^2}{P(z)} \frac{T_m}{P_m}.$$

This is a convex combination of the active literal row ratios. A minimizing row basis vector attains the minimum. The same argument with $\widetilde{P}_m$ proves (93). The two zero-row conclusions follow from (82). □

For a fixed distinguished vector z_* satisfying $\widetilde{P}(z_*) > 0$, put

$$\tau_{\text{rep},*}(X) := \frac{T(z_*)}{\widetilde{P}(z_*)}. \tag{94}$$

Theorem 6.3 (Exact distinguished weighted average)**.** *The distinguished repaired activation is exactly*

$$\boxed{\tau_{\text{rep},*}(X) = \sum_{m:\widetilde{P}_m>0} \frac{\widetilde{P}_m |z_{*,m}|^2}{\widetilde{P}(z_*)} g_m.} \tag{95}$$

The displayed coefficients are nonnegative and sum to one. If also $P(z_*) > 0$, then

$$\tau_{\text{hyp},*}(X) := \frac{T(z_*)}{P(z_*)} = \sum_{m:P_m>0} \frac{P_m |z_{*,m}|^2}{P(z_*)} \alpha_m \vartheta_m. \tag{96}$$

Neither distinguished identity implies the corresponding row minimum: a distinguished vector may avoid a bad row, whereas a full-row statement must test it.

Proof. Insert the definitions of $T(z_*)$, $\widetilde{P}(z_*)$, and g_m. The second formula is identical with P_m in place of $\widetilde{P}_m$. □

6.3 Sharp transfer through parent inflation

Theorem 6.4 (Inflation-corrected activation transfer)**.** *Assume $\mathfrak{I}_X < \infty$. For every z with $P(z) > 0$,*

$$\boxed{\frac{T(z)}{\widetilde{P}(z)} \geq \frac{1}{\mathfrak{I}_X} \frac{T(z)}{P(z)}.} \tag{97}$$

Consequently,

$$\boxed{\tau_{\mathrm{rep}} \geq \frac{\tau_{\mathrm{hyp}}}{\mathfrak{I}_X}.} \tag{98}$$

For every $a > 0$ and $L \geq 1$, a one-row terminal-faithful model attains simultaneously

$$\tau_{\mathrm{hyp}} = a, \qquad \mathfrak{I}_X = L, \qquad \tau_{\mathrm{rep}} = \frac{a}{L}. \tag{99}$$

Thus the factor $\mathfrak{I}_X^{-1}$ is sharp.

Proof. By theorem 5.1, $\widetilde{P}(z) \leq \mathfrak{I}_X P(z)$. A finite inflation constant excludes new-only rows, so $\widetilde{P}(z) > 0$ if and only if $P(z) > 0$, up to common zero rows. Multiplying the parent domination by the nonnegative number $T(z)/(P(z)\widetilde{P}(z))$ proves (97). Taking the infimum and using (90) gives (98).

For sharpness, take one literal terminal-active atom with $|c|^2 = 1$ and $|t_h|^2 = a$, and add terminal-null parent mass $L - 1$. Then

$$P_m = H_m = 1, \qquad T_m = a, \qquad \widetilde{P}_m = L,$$

which gives (99). □

If $\mathfrak{I}_X = +\infty$, there is no coefficient-uniform comparison from the literal parent to the repaired parent. Writing $\mathfrak{I}_X^{-1} = 0$ makes (98) formally true but quantitatively empty; it is not an activation theorem.

The results of this section are exact finite-dimensional identities. They do not prove that α_m, ϑ_m, g_m, or either activation constant has a favorable lower bound on the physical carrier. In particular, they contain no signed parity estimate and no prime-pair consequence.

7 A sharp threshold-capacity certificate

The rowwise minimax is the correct answer for a homogeneous full-coefficient theorem. A different exact certificate is available for one coefficient vector, or for a declared class on which a quantitative capacity lower bound is known. This section gives that certificate with its quantifiers exposed.

7.1 Good rows and occupied repaired-parent capacity

Use the repaired row masses $\widetilde{P}_m$ and the unchanged terminal masses T_m. Delete rows with $\widetilde{P}_m = 0$, and assume that the remaining set

$$\mathcal{M}_X^+ := \{m : \widetilde{P}_m > 0\} \tag{100}$$

is nonempty. Put

$$\widetilde{W}_X := \sum_{m \in \mathcal{M}_X^+} \widetilde{P}_m \tag{101}$$

and define the exact repaired-parent row activation

$$g_m := \frac{T_m}{\widetilde{P}_m} \geq 0 \qquad (m \in \mathcal{M}_X^+). \tag{102}$$

There is no assertion that $g_m \le 1$: the literal shaped terminal multiplier remains inside T_m.

For a threshold $\eta > 0$, define

$$\mathcal{G}_\eta := \{m \in \mathcal{M}_X^+ : g_m \ge \eta\}, \tag{103}$$

$$\delta_\eta := \frac{\sum_{m\in\mathcal{G}_\eta} \widetilde{P}_m}{\widetilde{W}_X} \in [0,1]. \tag{104}$$

Thus δ_η is a structural fraction measured in the exact repaired-parent denominator. It is not the unweighted density of good rows and is not an actual-coefficient energy fraction.

Let $C_X > 0$ be a declared coefficient bound and let z satisfy

$$|z_m| \le C_X \qquad (m \in \mathcal{M}_X^+). \tag{105}$$

Whenever $\widetilde{P}(z) > 0$, its occupied capacity is

$$\kappa_X(z; C_X) := \frac{\widetilde{P}(z)}{C_X^2 \widetilde{W}_X} \in (0,1]. \tag{106}$$

For a real number x, write $x_+ := \max\{x, 0\}$.

7.2 The exact threshold theorem

Theorem 7.1 (Sharp threshold-capacity lower bound)**.** *Fix $\eta > 0$ and $0 < \kappa_X \le 1$. If a coefficient vector z satisfies (105) and*

$$\widetilde{P}(z) \ge \kappa_X C_X^2 \widetilde{W}_X > 0, \tag{107}$$

then

$$\boxed{\frac{T(z)}{\widetilde{P}(z)} \ge \eta\left(1 - \frac{1-\delta_\eta}{\kappa_X}\right)_+ .} \tag{108}$$

More generally, if only $\delta_\eta \ge \underline{\delta}_\eta$ is known, then the same conclusion holds with $\underline{\delta}_\eta$ in place of δ_η.

Using only η, δ_η, the coefficient cap, and the capacity lower bound, the right side of (108) is optimal. In particular, it is positive exactly when

$$\boxed{\kappa_X > 1 - \delta_\eta.} \tag{109}$$

Proof. Write

$$E_{\mathrm{good}}(z) := \sum_{m\in\mathcal{G}_\eta} \widetilde{P}_m |z_m|^2, \qquad E_{\mathrm{bad}}(z) := \widetilde{P}(z) - E_{\mathrm{good}}(z). \tag{110}$$

The coefficient cap and (104) give

$$E_{\mathrm{bad}}(z) \le C_X^2 (1-\delta_\eta) \widetilde{W}_X.$$

Subtracting this from (107), and also using $E_{\mathrm{good}}(z) \ge 0$, yields

$$\frac{E_{\mathrm{good}}(z)}{\widetilde{P}(z)} \ge \left(1 - \frac{1-\delta_\eta}{\kappa_X}\right)_+ .$$

Since $T_m = g_m \widetilde{P}_m$ and $g_m \ge \eta$ on $\mathcal{G}_\eta$,

$$T(z) = \sum_m g_m \widetilde{P}_m |z_m|^2 \ge \eta E_{\mathrm{good}}(z),$$

which proves (108). Monotonicity in δ_η proves the version with a lower bound for that fraction.

For sharpness, normalize $C_X = \widetilde{W}_X = 1$ and prescribe $\delta \in [0,1]$, $\kappa \in (0,1]$. Take $g_m = \eta$ on a good part of parent capacity δ and $g_m = 0$ on its complement. If $0 < \delta < 1$ and $\kappa \le 1-\delta$, put zero coefficient energy on the good part and constant squared coefficient $\kappa/(1-\delta)$ on the bad part. The terminal quotient is zero. If $\kappa > 1-\delta$, saturate the bad part at squared coefficient one and put constant squared coefficient $(\kappa - 1 + \delta)/\delta$ on the good part. Then

$$\widetilde{P}(z) = \kappa, \qquad \frac{T(z)}{\widetilde{P}(z)} = \eta \frac{\kappa - 1 + \delta}{\kappa}.$$

These are equality cases for the two branches. When $\delta = 0$, take only bad rows; when $\delta = 1$, take only good rows. Constant squared coefficient κ gives the required occupied capacity at both endpoints. Finite weighted rows realize all of these models, so the bound cannot be improved from the stated data. $\square$

The theorem also has the exact actual-capacity form

$$\frac{T(z)}{\widetilde{P}(z)} \ge \eta \left(1 - \frac{1-\delta_\eta}{\kappa_X(z; C_X)}\right)_+ . \tag{111}$$

A certified lower bound for $\kappa_X(z; C_X)$ may then be substituted because the right side is increasing in the capacity variable.

7.3 Uniform and distinguished quantifiers

For $C_X > 0$ and $0 < \kappa_X \le 1$, let

$$\mathfrak{Z}_X(C_X, \kappa_X) := \left\{ z : \ |z_m| \le C_X, \quad \widetilde{P}(z) \ge \kappa_X C_X^2 \widetilde{W}_X \right\}. \tag{112}$$

Theorem 7.1 is coefficient-uniform on this declared restricted class. It is not a theorem on the entire bounded coefficient cube: that cube contains vectors with arbitrarily small occupied capacity and, whenever bad rows exist, vectors supported entirely on those rows.

Proposition 7.2 (The full-space boundary)**.** *On any homogeneous declared class $\mathfrak{Z}_X$ containing a nonzero multiple of each active row direction,*

$$\inf_{\substack{z \in \mathfrak{Z}_X \\ \widetilde{P}(z) > 0}} \frac{T(z)}{\widetilde{P}(z)} = \min_{m \in \mathcal{M}_X^+} g_m. \tag{113}$$

Consequently a favorable value of (108) for one distinguished physical vector does not imply a full-space uniform terminal lower bound.

Proof. Every quotient is a convex combination of the g_m, with weights $\widetilde{P}_m |z_m|^2 / \widetilde{P}(z)$. A permitted row direction attains the minimum. $\square$

For a distinguished vector $z_*(X)$, all of C_X, $\kappa_X(z_*; C_X)$, and the threshold η must concern that same vector and the same repaired parent. An averaged vector, an adaptively selected coefficient, or a capacity estimate computed with P_m instead of $\widetilde{P}_m$ cannot be inserted into (108).

7.4 What the certificate does not supply

The theorem is an implication from δ_η and occupied capacity. It proves neither input favorable. In particular:

(i) a positive number of terminal-active rows does not lower-bound their $\widetilde{P}_m$-capacity;

(ii) boundedness of the inherited coefficients does not imply $\kappa_X > 1 - \delta_\eta$;

(iii) positivity of each T_m does not give a fixed-power lower bound for g_m; and

(iv) the raw terminal form T still precedes label erasure, residual grouping, and the native physical reassembly map.

Thus the threshold theorem is a sharp L0 capacity law and, after the literal fixed-shift dictionary is verified, a conditional L1 certificate. It is not a new prime-distribution or cancellation estimate.

8 Sharp countermodels and zero tests

The exact formulas above separate identities from arithmetic lower bounds. This section shows that the separation is necessary. Every example is a finite coefficient-independent incidence with raw counting-measure norms. The examples are sharp models of the stated interfaces; they are not asserted to occur inside the literal fixed-shift prime carrier.

8.1 Faithful rectangles can have arbitrary parent inflation

Example 8.1 (Arbitrarily large faithful inflation)**.** *Take one block k, one row m, one cofactor r, one auxiliary label γ_0, and $N \geq 2$ base labels $j_0, \dots, j_{N-1}$. Let the declared ambient incidence contain*

$$u_i := (m, r, j_i, \gamma_0), \qquad 0 \leq i < N,$$

with $c_{u_i} = 1$. Let the literal pre-terminal incidence be

$$\mathcal{U}^{\mathrm{pre},+}_{X,k} = \{u_0\},$$

and set

$$t_h(u_0) = 1, \qquad t_h(u_i) = 0 \quad (1 \leq i < N).$$

The literal parent pair relation is the singleton

$$I_{\mathrm{pre},k} = \{(m, r)\}.$$

Its rectangle contains all N ambient points. It is parent-complete and terminal-atomic faithful because u_0 is the only terminal-active point. Nevertheless,

$$P_m = H_m = T_m = 1, \qquad \Delta_m = N - 1, \qquad \widetilde{P}_m = N. \tag{114}$$

Consequently,

$$\mathfrak{I}_X = N, \qquad \tau_{\mathrm{hyp}} = 1, \qquad \tau_{\mathrm{rep}} = \frac{1}{N}. \tag{115}$$

The terminal field and retained pair are unchanged; only the populated parent rectangle has grown.

Taking $N = 2$ and replacing the single terminal-null unit coefficient by one of modulus $\sqrt{L-1}$ realizes every real $L \geq 1$. Thus proposition 5.2 and theorem 6.4 are sharp at every admissible value of the constants.

Corollary 8.2 (One block and logarithmic label count give no inflation control)**.** *For every fixed $A > 0$, there is a family of examples with*

$$|K_X| = 1, \qquad \mathfrak{I}_X = X^{A+o(1)}. \tag{116}$$

Hence even the stronger statement $|K_X| = 1$, and therefore the usual $|K_X| = O(\log X) = X^{o(1)}$, supplies no upper bound on parent inflation.

Proof. In example 8.1, take $N = \lceil X^A \rceil$. All points have the same cofactor and hence the same external cofactor-block label. $\square$

This also proves that parent inflation is not a disguised block-count loss. A polynomial loss may occur entirely inside one block.

8.2 New-only rows and terminal-zero rows

Proposition 8.3 (New-only row no-go)**.** *There are terminal-faithful populated repairs with a row m_1 satisfying*

$$P_{m_1} = H_{m_1} = T_{m_1} = 0 < \widetilde{P}_{m_1}. \tag{117}$$

For every such repair,

$$\mathfrak{I}_X = +\infty, \qquad \tau_{\mathrm{rep}} = 0. \tag{118}$$

Thus no finite literal-to-repaired parent domination and no positive full-row repaired activation theorem can hold.

Proof. Take a literal terminal-active atom on a row m_0, with unit parent and terminal square. Add to the declared repair a different pair on a row m_1 whose populated rectangle contains one coordinate v with $c_v = 1$ and $t_h(v) = 0$, and declare v to be the only ambient coordinate over that added pair. The extra coordinate is terminal-null, so terminal faithfulness is preserved, but (117) holds with $\widetilde{P}_{m_1} = 1$. The row convention and theorems 5.1 and 6.2 give (118). Equivalently, the row direction e_{m_1} has positive repaired parent and zero terminal energy. $\square$

The canonical least parent-complete relation does not insert an arbitrary extra row. The point of the proposition is that any larger populated repair, change of row coordinates, or claimed comparison must prove that new-only rows are absent; terminal faithfulness alone does not prove it.

Proposition 8.4 (Terminal-zero row kill)**.** *If a required row satisfies*

$$P_{m_0} > 0, \qquad H_{m_0} = T_{m_0} = 0, \tag{119}$$

then

$$\tau_{\mathrm{hyp}} = \tau_{\mathrm{rep}} = 0 \tag{120}$$

for every parent-complete terminal-faithful repair on a full coefficient class containing e_{m_0}. Enlarging the retained pair relation cannot repair the zero without changing the terminal field.

Proof. The literal row direction has positive parent and zero terminal energy, so $\tau_{\mathrm{hyp}} = 0$. Terminal faithfulness leaves T_{m_0} unchanged, while parent completeness gives $\widetilde{P}_{m_0} \geq P_{m_0} > 0$. The same direction therefore forces $\tau_{\mathrm{rep}} = 0$. $\square$

A distinguished vector can avoid either bad row by having zero coefficient there. That is a different quantifier and yields only the weighted average in theorem 6.3.

8.3 The terminal-active-parent tautology

Example 8.5 (Deleting the alignment defect by redefining the parent)**.** *Fix $L \geq 1$. On one row, take two literal pre-terminal atoms:*

$$|c_{u_0}|^2 = 1, \quad t_h(u_0) = 1, \qquad |c_{u_1}|^2 = L - 1, \quad t_h(u_1) = 0,$$

omitting u_1 when $L = 1$. Then

$$P_m = L, \qquad H_m = T_m = 1, \qquad \frac{T_m}{P_m} = \frac{1}{L}. \tag{121}$$

If one replaces the literal parent by the terminal-active incidence $\{u_0\}$, the newly defined parent mass is $H_m = 1$, and the displayed ratio becomes

$$\frac{T_m}{H_m} = 1. \tag{122}$$

The apparent improvement can be arbitrarily large, but it is obtained by deleting the literal terminal-null parent mass $L-1$. It is not a lower bound relative to the original parent.

The example is the exact reason that multiplier intensity is not itself terminal activation. In the notation of theorem 6.1, redefining the parent as the terminal-active support silently replaces the alignment factor $\alpha_m = 1/L$ by 1.

8.4 Multiplier intensity is not a probability

Example 8.6 (Unbounded literal multiplier intensity)**.** *Take one literal terminal-active atom with $|c_u|^2 = 1$ and $|t_h(u)| = M$, where $M > 0$. Then*

$$P_m = H_m = 1, \qquad T_m = M^2, \qquad \vartheta_m = M^2. \tag{123}$$

The number M is arbitrary. Hence ϑ_m is a shaped multiplier intensity and can exceed 1 by an arbitrary factor.

Any argument which clips T_m/H_m at one, interprets it as a survival probability, or removes the literal shaped weight changes the physical quadratic form.

8.5 A largest block can be chosen only after positivity

For $z \in Z_X$, define the nonnegative shaped energy on block k by

$$T_k(z) := \sum_{m \in \mathcal{M}_X} T_{k,m} |z_m|^2. \tag{124}$$

Exact block disintegration gives

$$T(z) = \sum_{k \in K_X} T_k(z). \tag{125}$$

Theorem 8.7 (Largest-block quantifier and sharpness)**.** *Assume $1 \le L_X := |K_X| < \infty$. For each fixed vector z,*

$$\boxed{\max_{k \in K_X} T_k(z) \ge \frac{T(z)}{L_X}.} \tag{126}$$

The constant L_X^{-1} is sharp. Consequently:

(i) *if a prior theorem gives*

$$T(z) \ge X^{-\lambda + o(1)} P(z)$$

for one fixed distinguished z, and $L_X = O(\log X)$, then some block $k = k(z, X)$ satisfies the same fixed-power lower bound up to $X^{o(1)}$;

(ii) *if $T(z) = 0$, every block energy is zero, so the largest-block step creates no activation;*

(iii) *the maximizing block may depend on the row or coefficient vector, and (126) alone does not select one block supporting a coefficient-uniform lower bound.*

Proof. The maximum of L_X nonnegative numbers is at least their average. Equality holds when all $T_k(z) = T(z)/L_X$, proving sharpness. The first consequence uses $L_X^{-1} = X^{-o(1)}$; the second is immediate from nonnegativity.

For the last assertion, take two rows and two blocks. Put

$$P_{m_1} = P_{m_2} = 1, \qquad T_{1,m_1} = 1, \quad T_{2,m_2} = 1,$$

with the two cross terms zero. The total terminal form equals the parent form, so its uniform activation constant is one. On block 1, the row direction e_{m_2} has zero energy; on block 2, the direction e_{m_1} has zero energy. Thus neither single block has a positive full-row activation constant. $\square$

The block theorem is therefore downstream of activation, not a source of activation. A logarithmic number of blocks costs no fixed power only after a positive total terminal estimate has been proved in the same coefficient scope.

8.6 Claim boundary

These countermodels prove non-implications for finite parent and terminal interfaces. They show that terminal faithfulness does not control parent inflation, that label count does not control populated parent mass, that zero rows kill uniform activation, and that block selection cannot create positivity. They do not show that the literal arithmetic carrier has a bad row or large inflation. Conversely, none of the exact identities in this paper proves positive physical activation, signed Möbius cancellation, a fixed-shift parity estimate, or a prime-pair lower bound.

9 Denominator compatibility and the effective activation loss

A terminal numerator becomes useful only when every paper in the chain uses the same parent denominator. This section records the exact compatibility requirements, defines the inflation and effective terminal losses, and inserts them into the inherited endpoint ledger without double charging.

9.1 The literal compatibility audit

Fix either a declared coefficient class $\mathfrak{Z}_X$ or one distinguished singleton $\{z_*(X)\}$. Every item below must concern that same scope.

Definition 9.1 (Compatible realization). A realization of the TPC–47/51/54/59/62/63 chain is *denominator compatible* if all of the following hold [1–6].

(D1) The TPC–47 physical slice uses one fixed h_0, its literal row coefficient z_m, one cofactor sign $\mu(r)$, and the actual coefficient-independent full incidence. No frozen shift or independently selected coefficient vector replaces this slice.

(D2) The TPC–51 direct sum retains the same (m, r, j, γ)-atoms and their raw counting-measure norm. A Mellin L^1 envelope, a quotient-frame norm, or a representation-dependent layer norm is not substituted for the physical parent.

(D3) The TPC–54 capacity weights are exactly the row masses of the parent used here. After a populated repair they are $\widetilde{P}_m$, so both δ_η and κ_X are computed with $\widetilde{P}_m$, not with source-row cardinality, P_m, or a terminal-active submeasure.

(D4) The TPC–59 terminal multiplier and support predicate give exactly T_m on the same atoms. The compatible pre-terminal denominator is not replaced by the terminal-support mass H_m.

(D5) The TPC–63 provenance test is applied before pair rectangularization. Pair-bad blocks retain their literal active incidence; on pair-good blocks every added parent atom and its inflation remain in $\widetilde{P}_m$. Terminal-active support is not used retroactively as the parent.

(D6) In the TPC–62 chain, the form leaving this activation module is literally the form entering the represented/native module. Every external block label is retained, or its actual erasure map and range-restricted lower bound are inserted as a separate reassembly arrow.

These conditions are identities of objects, not estimates. When they hold, the relevant beginning of the chain is

$$\boxed{P(z) \xrightarrow[\text{parent completion}]{\widetilde{P}=P+\Delta} \widetilde{P}(z) \xrightarrow[\text{literal terminal map}]{} T(z).} \tag{127}$$

The first arrow is not a lower transfer: it records a possible denominator enlargement. The lower constant for the second arrow must therefore be measured by $T(z)/\widetilde{P}(z)$.

9.2 Exact constants and exponent definitions

Assume first that the declared class contains at least one vector with positive literal parent. Define

$$\mathfrak{I}_X(\mathfrak{Z}_X) := \sup_{\substack{z\in\mathfrak{Z}_X\\ \widetilde{P}(z)>0}} \frac{\widetilde{P}(z)}{P(z)} \in [1,+\infty], \tag{128}$$

$$c^{\text{lit}}_{\text{term}}(X;\mathfrak{Z}_X) := \inf_{\substack{z\in\mathfrak{Z}_X\\ P(z)>0}} \frac{T(z)}{P(z)}, \tag{129}$$

$$c^{\text{eff}}_{\text{term}}(X;\mathfrak{Z}_X) := \inf_{\substack{z\in\mathfrak{Z}_X\\ \widetilde{P}(z)>0}} \frac{T(z)}{\widetilde{P}(z)}. \tag{130}$$

In (128), the quotient is understood as $+\infty$ when $P(z) = 0 < \widetilde{P}(z)$. Thus a finite compatible inflation constant automatically excludes a new-only vector from the declared scope. For the canonical hybrid repair there is no new-only row, and the full-space value of $\mathfrak{I}_X$ is the maximum row inflation. For a distinguished singleton, all three quantities are ratios on that one vector and must retain the distinguished label.

The exact comparison is

$$\boxed{c^{\text{eff}}_{\text{term}}(X;\mathfrak{Z}_X) \geq \frac{c^{\text{lit}}_{\text{term}}(X;\mathfrak{Z}_X)}{\mathfrak{I}_X(\mathfrak{Z}_X)}} \tag{131}$$

whenever the right side is defined with finite $\mathfrak{I}_X$. Indeed, finiteness gives $P(z) > 0$ whenever $\widetilde{P}(z) > 0$, and for every such vector

$$\frac{T(z)}{\widetilde{P}(z)} = \frac{T(z)}{P(z)} \left(\frac{\widetilde{P}(z)}{P(z)}\right)^{-1} \geq \frac{c^{\text{lit}}_{\text{term}}(X;\mathfrak{Z}_X)}{\mathfrak{I}_X(\mathfrak{Z}_X)}.$$

Taking the infimum proves (131). This is sharp already in a one-row model. A direct estimate relative to $\widetilde{P}$, such as theorem 7.1, can be stronger.

For an asymptotic family of scopes, define the optimal fixed-power losses

$$\lambda_{\text{infl}} := \limsup_{X\to\infty} \frac{\log \mathfrak{I}_X(\mathfrak{Z}_X)}{\log X}, \tag{132}$$

$$\lambda^{\text{eff}}_{\text{term}} := \limsup_{X\to\infty} \frac{\log^+(1/c^{\text{eff}}_{\text{term}}(X;\mathfrak{Z}_X))}{\log X}, \tag{133}$$

provided the corresponding constant is positive eventually. If $\mathfrak{I}_X = +\infty$ or $c^{\mathrm{eff}}_{\mathrm{term}} = 0$ infinitely often, assign the relevant loss $+\infty$. Here $\log^+ u := \max\{\log u, 0\}$. Define $\lambda^{\mathrm{lit}}_{\mathrm{term}}$ analogously from $c^{\mathrm{lit}}_{\mathrm{term}}$.

Proposition 9.2 (Effective loss from the two honest routes)**.** *In one fixed nonempty quantifier scope. In route* (ii) *this scope is a normalized or capacity-restricted slice and need not be homogeneous:*

(i) *if* $\mathfrak{I}_X \le X^{\lambda_{\mathrm{infl}}+o(1)}$ *and* $c^{\mathrm{lit}}_{\mathrm{term}} \ge X^{-\lambda^{\mathrm{lit}}_{\mathrm{term}}+o(1)}$, *then*

$$c^{\mathrm{eff}}_{\mathrm{term}} \ge X^{-(\lambda^{\mathrm{lit}}_{\mathrm{term}}+\lambda_{\mathrm{infl}})+o(1)}; \tag{134}$$

(ii) *if common* $C_X > 0$ *and* $0 < \kappa_X \le 1$ *exist such that every* $z \in \mathfrak{Z}_X$ *satisfies*

$$|z_m| \le C_X \quad (m \in \mathcal{M}^+_X), \qquad \widetilde{P}(z) \ge \kappa_X C_X^2 \widetilde{W}_X > 0,$$

and the threshold-capacity inputs give

$$\eta_X \left(1 - \frac{1-\delta_{\eta_X}}{\kappa_X}\right)_+ \ge X^{-\lambda_{\mathrm{cap}}+o(1)}, \tag{135}$$

on the same declared class, then

$$c^{\mathrm{eff}}_{\mathrm{term}} \ge X^{-\lambda_{\mathrm{cap}}+o(1)}. \tag{136}$$

If both routes apply, the larger lower constant may be used. In particular,

$$\lambda^{\mathrm{eff}}_{\mathrm{term}} \le \min\{\lambda^{\mathrm{lit}}_{\mathrm{term}} + \lambda_{\mathrm{infl}}, \lambda_{\mathrm{cap}}\}. \tag{137}$$

No equality is asserted: a direct terminal estimate may be better than either certificate.

Proof. The first conclusion is (131) in exponent form. The second is theorem 7.1. Taking the better of two proved lower bounds gives (137). ☐

If $\mathfrak{Z}_X = \{z_*(X)\}$, every constant and exponent above is distinguished. A bound proved there cannot be inserted into the full-space versions of (128)–(130).

9.3 The compatible physical chain

Let $\mathcal{Q}_{\mathrm{in}}$, $\mathcal{Q}_A$, $\mathcal{Q}_F$, and $\mathcal{Q}_{\mathrm{out}}$ denote, respectively, the declared upstream input, represented native output, full represented-plus-missing output, and downstream reconnected output. The terminal form T and repaired parent $\widetilde{P}$ are inserted as two additional named nodes.

Theorem 9.3 (Exact five-arrow compatible chain)**.** *Suppose all six forms are nonnegative quadratic forms evaluated on the same initial* z *in the same scope. Suppose also that*

$$c_{\mathrm{pre}}, c_{\mathrm{term}}, c_{\mathrm{post}}, c_{\mathrm{reasm}}, c_{\mathrm{out}} \ge 0$$

and that

$$\widetilde{P}(z) \ge c_{\mathrm{pre}}(X)\mathcal{Q}_{\mathrm{in}}(z), \tag{138}$$
$$T(z) \ge c_{\mathrm{term}}(X)\widetilde{P}(z), \tag{139}$$
$$\mathcal{Q}_A(z) \ge c_{\mathrm{post}}(X)T(z), \tag{140}$$
$$\mathcal{Q}_F(z) \ge c_{\mathrm{reasm}}(X)\mathcal{Q}_A(z), \tag{141}$$
$$\mathcal{Q}_{\mathrm{out}}(z) \ge c_{\mathrm{out}}(X)\mathcal{Q}_F(z). \tag{142}$$

Every output carrier and normalization in one line is required to be the literal input carrier and normalization in the next. Then

$$\boxed{\mathcal{Q}_{\mathrm{out}}(z) \geq c_{\mathrm{out}} c_{\mathrm{reasm}} c_{\mathrm{post}} c_{\mathrm{term}} c_{\mathrm{pre}}\, \mathcal{Q}_{\mathrm{in}}(z).} \tag{143}$$

The statement is uniform only if all five inequalities and all their remainders are uniform on the same $\mathfrak{Z}_X$.

Proof. Apply the five inequalities successively to the named intermediate forms generated by the same z. No supremum, infimum, selector, shift, or coefficient vector changes during the substitutions. □

The exact TPC–47/51 dictionary may make $c_{\mathrm{pre}} = 1$ on a literally identical carrier. TPC–54 selection or TPC–59 occupancy costs belong to c_{pre} only if they compare $\mathcal{Q}_{\mathrm{in}}$ with the precise repaired parent used in (139). TPC–62 reassembly begins only at (141); it cannot repair a zero terminal constant.

9.4 No-double-charge ledger and stop rule

Write

$$\Lambda_{\mathrm{pre}} := \lambda_{\mathrm{sel}} + \lambda_{\mathrm{occ}} + \lambda_{\mathrm{blk}}. \tag{144}$$

When each symbol corresponds to a genuinely distinct arrow in theorem 9.3, the inherited physical ledger is

$$\boxed{\begin{aligned}\Lambda_{\mathrm{tot}}^{(64)} := \lambda_{\mathrm{sel}} + \lambda_{\mathrm{occ}} + \lambda_{\mathrm{blk}} + \lambda_{\mathrm{term}}^{\mathrm{eff}} + \lambda_{\mathrm{sing}} \\ + \lambda_H + \lambda_{\mathrm{aff}} + \lambda_{\mathrm{reasm}} + \lambda_{\mathrm{bdry}} + \lambda_{\mathrm{tail}}.\end{aligned}} \tag{145}$$

The strict endpoint test is

$$\boxed{\Lambda_{\mathrm{tot}}^{(64)} < \frac{1}{400}.} \tag{146}$$

The following no-double-charge rules are part of the formula.

(a) If $\lambda_{\mathrm{term}}^{\mathrm{eff}}$ is obtained from $\lambda_{\mathrm{term}}^{\mathrm{lit}} + \lambda_{\mathrm{infl}}$, only $\lambda_{\mathrm{term}}^{\mathrm{eff}}$ enters (145); the two diagnostic inputs are not added again.

(b) If the capacity theorem proves the terminal arrow directly relative to $\widetilde{P}$, then $\eta_X, \delta_{\eta_X}, \kappa_X$, and λ_{cap} are ingredients of the one term $\lambda_{\mathrm{term}}^{\mathrm{eff}}$, not four further arrows.

(c) If selection, occupancy, or block losses were already used to prove the same direct ratio $T/\widetilde{P}$, they must be removed from Λ_{pre}. Conversely, if they compare an earlier $\mathcal{Q}_{\mathrm{in}}$ with $\widetilde{P}$, they remain separate.

(d) A represented-reference exponent used only to certify λ_{reasm} is not charged again if its physical arrow is already included in $\lambda_H + \lambda_{\mathrm{aff}}$.

(e) Exact set identities, faithful hypergraph retention, literal coordinate permutations or unitary relabelings, and $O(\log X)$ diagnostic labels have zero fixed-power cost. A more general coordinate change has zero such cost only after a uniform or $X^{o(1)}$ condition-number bound is proved; otherwise its loss must enter the ledger. Erasing diagnostic labels is not a coordinate change and belongs to λ_{reasm}.

If a mandatory constant in theorem 9.3 is zero infinitely often, the matching exponent route stops. If a necessary proved loss is at least $1/400$, the exponent-only endpoint route also stops, including at equality. A separate constant-sensitive endpoint argument at equality would lie outside this strict exponent ledger and would not satisfy (146). A distinguished success, an averaged-shift success, or a different-denominator success does not evade either stop rule.

10 Result ledger, claim boundary, and the next arithmetic target

TPC–64 separates true terminal activation from two representation effects: deletion of parent mass and inflation by terminal-null crossed coordinates. It supplies exact finite-scale tests and a sharp capacity certificate. It does not prove that the physical terminal incidence is large.

10.1 Result ledger

Object	Exact conclusion	Status
Literal parent and terminal	The full incidence gives exact row masses P_m, H_m, T_m, raw block disintegration, and, on rows with $P_m, H_m > 0$, the factorization $T_m/P_m = (H_m/P_m)(T_m/H_m)$.	L0 theorem; L1 dictionary
Uniform activation	On a class containing every active row direction, the optimal constant is $\min_{P_m>0} T_m/P_m$; one zero row kills it.	L0 minimax
Parent-faithful pair lift	A pair lift exists exactly when the parent projection creates no terminal-active spurious cross. The least faithful relation and the hypergraph alternative are explicit.	L0 iff test
Parent inflation	A parent-complete, terminal-faithful populated rectangle has $\widetilde{P}_m = P_m + \Delta_m$ while preserving T_m. The optimal domination is the largest row inflation; a new-only row makes it infinite.	L0 theorem
Canonical hybrid	Pair-good blocks use their least parent-complete lift, pair-bad blocks retain the literal active incidence, and the literal-active-hypergraph baseline has exactly zero inflation.	L0 construction
Threshold capacity	If $g_m = T_m/\widetilde{P}_m$, then threshold capacity δ_η and occupied capacity κ_X give the sharp lower bound $\eta(1-(1-\delta_\eta)/\kappa_X)_+$.	L0 sharp theorem
Denominator chain	Literal activation, inflation, represented transfer, reassembly, and downstream arrows compose only with the same vector, carrier, raw norm, and quantifier scope.	L0 compatibility
Endpoint	Inflation is charged once inside $\lambda^{\mathrm{eff}}_{\mathrm{term}}$; all distinct physical losses must sum strictly below $1/400$.	L0 ledger; L1 conditional

10.2 The L0/L1/L2 boundary

Level	Established here	Not licensed by that level
L0	Exact finite-incidence parent/terminal forms, faithful-lift criteria, inflation identities, rowwise minimax laws, capacity extremizers, and compatible-form composition.	No counting, density, or lower-bound assertion for the declared prime candidates, and no assertion that any asymptotic constant is favorable.
L1	Conditional specialization to one fixed nonzero h_0, the literal TPC coefficient field, one cofactor Möbius sign, raw atomic norms, and the stated terminal predicate.	No proof that every physical row activates, that δ_η or κ_X passes the threshold, or that physical label erasure is bounded below.
L2	None.	No signed fixed-shift cancellation estimate, no specialization to $h_0 = 2$, no parity-barrier advance, no prime-pair lower bound, and no twin-prime consequence.

10.3 What is not proved

The following statements remain absent.

1. No favorable lower bound is proved for δ_{η_X} at any fixed or polynomially shrinking threshold η_X.
2. The actual physical coefficient vector is not shown to satisfy $\kappa_X > 1 - \delta_{\eta_X}$, let alone with an endpoint-quality quantitative margin.
3. No bound
$$\min_{m:P_m>0} \frac{T_m}{P_m} \geq X^{-\lambda+o(1)}$$
is proved for the full coefficient class. Finite-scale positivity of every row would still be weaker than such a bound.
4. No subpower or endpoint-quality bound is proved for the parent inflation of a pair-populated repair. The literal-active-hypergraph baseline avoids that inflation but may not satisfy a downstream demand for pair coordinates.
5. The labeled terminal square is not shown to survive dyadic-label erasure, residual grouping, comparison with the native affine Gram, boundary reconnection, or the ultra-tail map.
6. None of the nonnegative endpoint losses is newly shown to fit inside the strict total budget.

The capacity theorem is therefore not evidence that δ_η or κ_X is favorable. Likewise, failure of its threshold is failure of that certificate; it does not prove that the actual terminal ratio $T(z)/\widetilde{P}(z)$ is small.

10.4 The next genuinely arithmetic theorem

The next layer must estimate the terminal incidence itself. There are two honest quantifier scopes.

Uniform row target. For a full-coefficient theorem, prove on the literal multiscale carrier

$$\boxed{T_m \geq X^{-\lambda_{\mathrm{row}}+o(1)}\widetilde{P}_m \quad \text{for every } m \text{ with } \widetilde{P}_m > 0,} \tag{147}$$

with one uniform remainder and an exponent compatible with (146). Expanded literally, this is a weighted lower theorem for terminal candidates

$$p = \frac{mj + h_0}{r} \tag{148}$$

which pass the declared prime, cofactor, squarefree, coprimality, mask, and profile tests on every active source row. A positive average over rows does not prove (147).

Distinguished capacity target. For the actual physical vector $z_*(X)$, one may instead prove, for a declared $\eta_X > 0$,

$$\delta_{\eta_X} = \frac{\sum_{m:g_m \geq \eta_X} \widetilde{P}_m}{\sum_m \widetilde{P}_m}, \tag{149}$$

$$\kappa_X(z_*; C_X) = \frac{\widetilde{P}(z_*)}{C_X^2 \sum_m \widetilde{P}_m}, \tag{150}$$

and establish the quantitative threshold

$$\boxed{\eta_X \left(1 - \frac{1-\delta_{\eta_X}}{\kappa_X(z_*; C_X)}\right)_+ \geq X^{-\lambda^{\mathrm{eff}}_{\mathrm{term},*}+o(1)}.} \tag{151}$$

This gives a distinguished terminal arrow only. It cannot be entered as the uniform constant in (147).

The natural arithmetic work is therefore a weighted prime-candidate census in the progressions $mj + h_0 \equiv 0 \pmod r$, simultaneously with the literal row, orbit, cofactor, and terminal windows. A successful lower theorem advances the L1 activation gate. An explicit active row with $T_m = 0$, or a proved endpoint-sized dilution, closes the corresponding uniform architecture and is an honest negative result.

After activation. Only after one of the two targets above is proved should the next paper seek a represented-reference Gram lower bound on precisely the same terminal carrier. Determinant-pair cancellation, physical reassembly, boundary terms, and the tail remain later gates.

Conclusion. TPC–64 gives the exact meaning of true terminal activation relative to a parent-faithful denominator. Pair completion can preserve the numerator while diluting that denominator; threshold capacity can recover a sharp lower bound only when its two inputs pass a strict quantitative test. Neither mechanism supplies the missing arithmetic incidence theorem. The next progress must come from the literal terminal candidates, not from a relabeling of their parent.

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